

Relative Motion 1

Linking Concepts in Pre-Uni Physics 5.1

Subject & topics: Physics | Mechanics | Kinematics Stage & difficulty: A Level P1

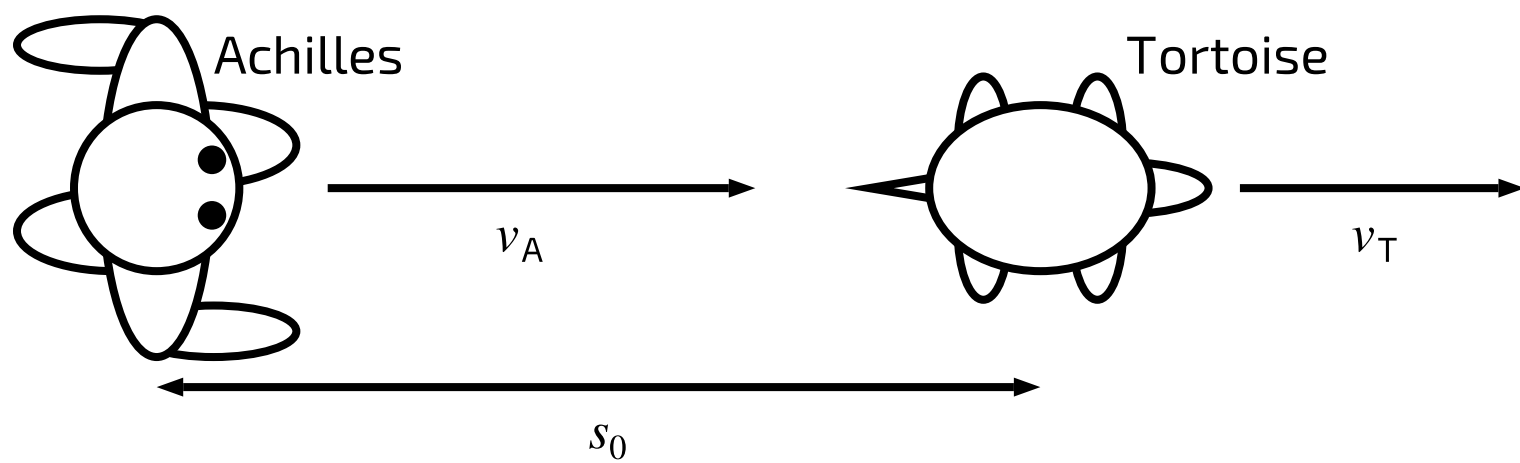


Figure 1: Achilles chasing the tortoise.

Quantities:

v_A velocity of Achilles (m s^{-1})

v_T velocity of tortoise (m s^{-1})

T time for Achilles to catch up (s)

s_0 initial displacement (m)

s displacement (m)

t time since start (s)

Equations:

$$v = \frac{s}{t}$$

Use the equations above to derive expressions for:

Part A

The velocity of Achilles relative to the tortoise

the velocity of Achilles relative to the tortoise v_{REL} .

The following symbols may be useful: T , s , s_0 , t , v_A , v_{REL} , v_T

Part B

The time for Achilles to catch up

the time for Achilles to catch up with the tortoise T , in terms of v_A and v_T .

The following symbols may be useful: T , s , s_0 , t , v_A , v_{REL} , v_T

Part C

The displacement of the tortoise relative to Achilles

the displacement of the tortoise s relative to Achilles as a function of time t .

The following symbols may be useful: T , s , s_0 , t , v_A , v_{REL} , v_T

Question deck:

STEM SMART Physics 26 - Collisions

Relative Motion 3

Linking Concepts in Pre-Uni Physics 5.3

Subject & topics: Physics | Mechanics | Kinematics Stage & difficulty: A Level P1

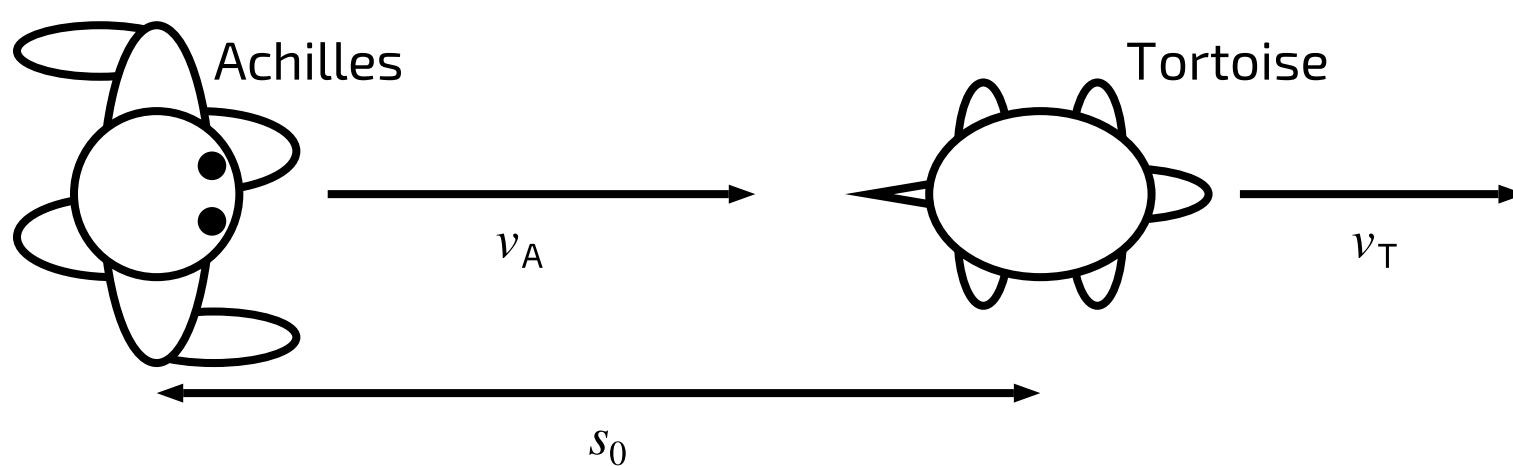


Figure 1: Achilles chasing the tortoise.

Following on from [Example 1](#), when the tortoise travelling at 18.0 m s^{-1} is 1.00 km away from Achilles, Achilles gets into a motor vehicle that can travel at 96.5 km h^{-1} . Calculate how far ahead of the tortoise Achilles is after 2 minutes.

Question deck:

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Conservation of Momentum 5

Essential Pre-Uni Physics F2.5

Subject & topics: Physics | Mechanics | Dynamics

Stage & difficulty: GCSE C3, A Level C2

A rocket (containing a space probe) is travelling at 7000 m s^{-1} in outer space. The 2000 kg probe is ejected from the front of the rocket (forwards) using a big spring. If the speed of the probe afterwards is 7200 m s^{-1} , and the rest of the rocket has a mass of 6000 kg , what is the speed of the rest of the rocket? Give your answer to 4 significant figures.

Question deck:

STEM SMART Physics 26 - Collisions

Elastic Collisions 1

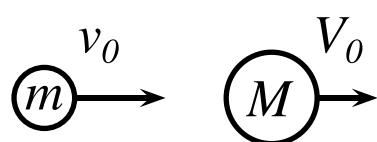
Linking Concepts in Pre-Uni Physics 4.1

Subject & topics: Physics | Mechanics | Dynamics Stage & difficulty: A Level P1



This question has been amended and may differ from the version you have in your book (if you are using a printed book). The original question can be found [by clicking here](#).

Before collision



After collision



Figure 1: Definitions of variables used in elastic collisions questions

Quantities:

p, P momentum (kg m s^{-1})

v, V velocity (m s^{-1})

k, K kinetic energy (J)

m, M mass (kg)

Equations:

$$p = mv \qquad k = \frac{1}{2}mv^2 \qquad P = MV \qquad K = \frac{1}{2}MV^2$$

$$p_0 + P_0 = p_1 + P_1 \qquad k_0 + K_0 = k_1 + K_1$$

Use the equations above to derive expressions for:

Part A

Final velocity V_1 of M

the final velocity V_1 of M if M was stationary at the beginning and the initial and final velocities of m (v_0 and v_1) are known.

The following symbols may be useful: M , V_1 , m , v_0 , v_1

Part B

V_1 with equal masses and m stopped

V_1 if the masses are equal ($M = m$), M begins at rest ($V_0 = 0$), m is stopped by the collision ($v_1 = 0$) and v_0 is known.

The following symbols may be useful: M , V_0 , V_1 , m , v_0 , v_1

Part C

Separation speed in terms of approach speed

the separation speed $V_1 - v_1$ in terms of the approach speed $v_0 - V_0$.

Hint: Start by writing out $k_0 - k_1 = K_1 - K_0$ in terms of m , M , v_0 , v_1 , V_0 and V_1 . Factorise your expression, then simplify it remembering that $p_0 - p_1 = P_1 - P_0$.

The following symbols may be useful: V_0 , V_1 , v_0 , v_1

Question deck:

STEM SMART Physics 26 - Collisions

Elastic Collisions 6

Linking Concepts in Pre-Uni Physics 4.6

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** A Level C1

In space, an elastic 'sling shot' collision is arranged between a $6.4 \times 10^{24} \text{ kg}$ planet moving at 9.0 km s^{-1} towards a 6000 kg spacecraft which is also moving at 4.5 km s^{-1} towards the planet.

Calculate the final speed of the spacecraft.

Question deck:

STEM SMART Physics 26 - Collisions

Elastic Collisions 8

Linking Concepts in Pre-Uni Physics 4.8

Subject & topics: Physics | Mechanics | Dynamics Stage & difficulty: A Level P1

A neutron (of mass m) travelling at $2.4 \times 10^5 \text{ m s}^{-1}$ collides elastically with a stationary iron nucleus (mass $M = 65m$) head on. Calculate,

Part A

The final speed of the nucleus

Calculate the final speed of the iron nucleus.

Part B

The percentage of the energy given to the nucleus

Calculate the percentage of the neutron's kinetic energy which is given to the nucleus.

Question deck:

STEM SMART Physics 26 - Collisions

Neutron Moderator

100 Challenge Problems in Pre-Uni Physics Q78

Subject & topics: Physics | Waves & Particles | Nuclear

Stage & difficulty: A Level C2

A fast moving neutron produced from a nuclear reaction undergoes collisions with initially stationary moderator particles.

A neutron undergoes three successive head-on elastic collisions with identical stationary moderator particles. Its speed is reduced to 60.6% of its initial value.

Find the mass of one moderator particle.

Created for isaacscience.org by Kimlam Nguyen

Question deck:

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